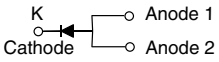
High Current Density Surface-Mount High Voltage Schottky Rectifier

eSMP® Series



SMPC (TO-277A)



LINKS TO ADDITIONAL RESOURCES

3D Models | Marking | Packages

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	12 A
V_{RRM}	100 V
I_{FSM}	215 A
V_F at I_F = 12 A	0.66 V
T_J max.	175 °C
Package	SMPC (TO-277A)
Circuit configuration	Single

FEATURES

AUTOMOTIVE GRADE Available

RoHS COMPLIANT HALOGEN FREE

- Very low profile - typical height of 1.1 mm
- Ideal for automated placement
- Guardring for overvoltage protection
- High barrier technology, T_J = 175 °C maximum
- Low leakage current
- Enhanced for high surge endurance
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- AEC-Q101 qualified available
- Automotive ordering code: base P/NHM3
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

TYPICAL APPLICATIONS

For use in high frequency rectifier of switching mode power supplies, freewheeling diodes, DC/DC converters, or polarity protection application.

MECHANICAL DATA

Case: SMPC (TO-277A)
Molding compound meets UL 94 V-0 flammability rating
Base P/N-M3 - halogen-free, RoHS-compliant, and commercial grade
Base P/NHM3 - halogen-free, RoHS-compliant, and AEC-Q101 qualified

Terminals: matte tin plated leads, solderable per J-STD-002 and JESD 22-B102
M3 suffix meets JESD 201 class 2 whisker test, HM3 suffix meets JESD 201 class 2 whisker test

MAXIMUM RATINGS (T_A = 25 °C unless otherwise noted)

PARAMETER		SYMBOL	SS12PH102	UNIT
Device marking code			12H102	
Maximum repetitive peak reverse voltage		V_{RRM}	100	V
Maximum average forward rectified current (fig. 1)		$I_{F(AV)}$ (1)	12	A
		$I_{F(AV)}$ (2)	3.4	
Non-repetitive peak forward surge current	8.3 ms half sine-wave superimposed on rated load	I_{FSM}	215	A
	100 µs square pulse		950	
Operating junction temperature range		T_J (3)	-40 to +175	°C
Storage temperature range		T_{STG}	-55 to +175	°C

Notes

(1) Mounted on 30 mm x 30 mm pad areas aluminum PCB
(2) Free air, mounted on recommended copper pad area, 2 oz, FR4 PCB
(3) The heat generated must be less than the thermal conductivity from junction-to-ambient: $dP_D/dT_J < 1/R_{\theta JA}$

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ELECTRICAL CHARACTERISTICS (T_A = 25 °C unless otherwise noted)						
PARAMETER	**TEST CONDITIONS**		**SYMBOL**	**TYP.**	**MAX.**	**UNIT**
Instantaneous forward voltage	I_F = 6 A	T_A = 25 °C	V_F (1)	0.73	-	V
	I_F = 12 A			0.81	0.85	
	I_F = 6 A	T_A = 125 °C		0.59	-	
	I_F = 12 A			0.66	0.71	
Reverse current	V_R = 70 V	T_A = 25 °C	I_R (2)	0.00003	-	mA
		T_A = 125 °C		0.15	-	
	V_R = 100 V	T_A = 25 °C		-	0.003	
		T_A = 125 °C		0.3	0.6	
Typical junction capacitance	4.0 V, 1 MHz		C_J	240	-	pF

Notes

(1) Pulse test: 300 µs pulse width, 1 % duty cycle

(2) Pulse test: Pulse width ≤ 5 ms

THERMAL CHARACTERISTICS (T_A = 25 °C unless otherwise specified)			
PARAMETER	**SYMBOL**	**SS12PH102**	**UNIT**
Typical thermal resistance	$R_{\theta JA}$ (1)(2)	75	°C/W
	$R_{\theta JM}$ (3)	4	

Notes

(1) The heat generated must be less than the thermal conductivity from junction to ambient: $dP_D/dT_J < 1/R_{\theta JA}$

(2) Free air mounted on recommended copper pad area; thermal resistance $R_{\theta JA}$ - junction to ambient

(3) Mounted on 30 mm x 30 mm aluminum PCB; thermal resistance $R_{\theta JM}$ - junction to mount

ORDERING INFORMATION TABLE

Device code

SS	12	P	H	10	2	H	M3
1	2	3	4	5	6	7	8

1 - Vishay planar Schottky product

2 - Current rating (12 = 12 A)

3 - Package type (P = SMPC (TO-277A))

4 - Process type option (H = low I_R)

5 - Voltage rating (10 = 100 V)

6 - Planar Schottky generation option (2 = gen 2)

7 - Quality grade (H = AEC-Q101 qualified, - = industry grade)

8 - Material / Environment category
(M3 = halogen-free, RoHS-compliant, and termination lead (Pb)-free)

ORDERING INFORMATION (Example)				
PREFERRED P/N	**UNIT WEIGHT (g)**	**PACKAGE CODE**	**BASE QUANTITY**	**DELIVERY MODE**
SS12PH102-M3/H	0.10	H	1500	7" diameter plastic tape and reel
SS12PH102-M3/I	0.10	I	6500	13" diameter plastic tape and reel
SS12PH102HM3/H (1)	0.10	H	1500	7" diameter plastic tape and reel
SS12PH102HM3/I (1)	0.10	I	6500	13" diameter plastic tape and reel

Note

(1) AEC-Q101 qualified

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RATINGS AND CHARACTERISTICS CURVES (T_A = 25 °C unless otherwise noted)

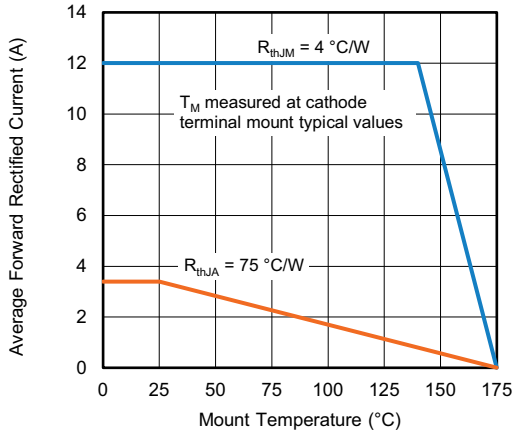


Fig. 1 - Maximum Forward Current Derating Curve

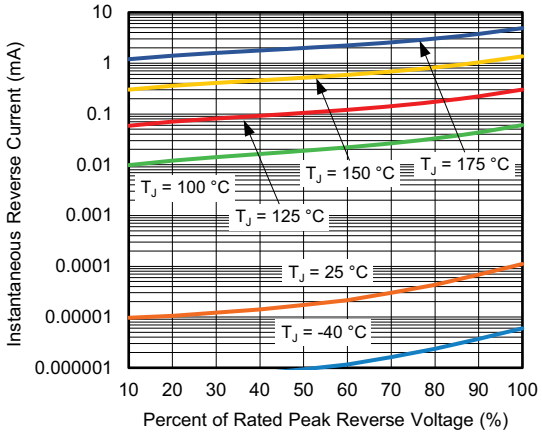


Fig. 4 - Typical Reverse Characteristics

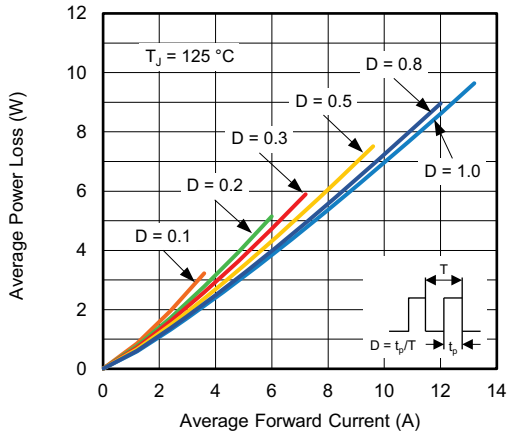


Fig. 2 - Forward Power Loss Characteristics

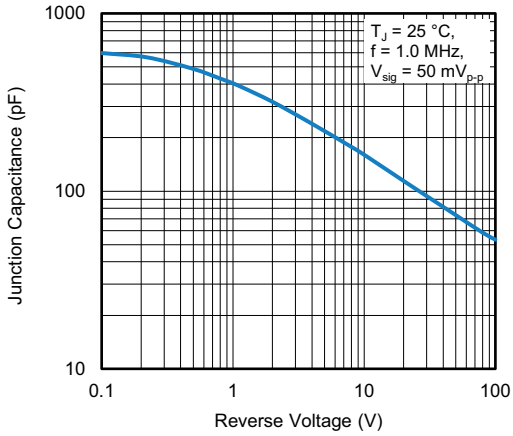


Fig. 5 - Typical Junction Capacitance

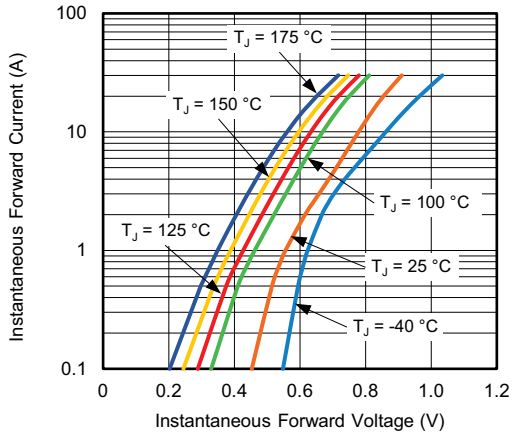


Fig. 3 - Typical Instantaneous Forward Characteristics

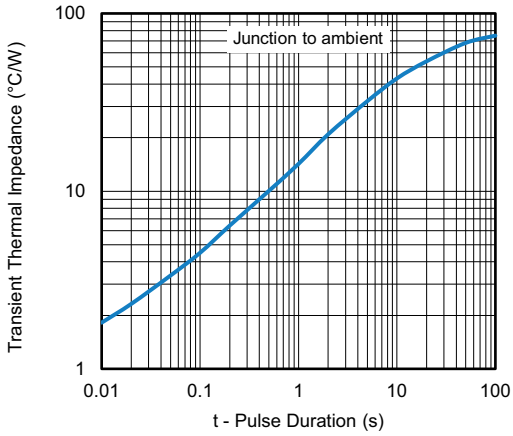
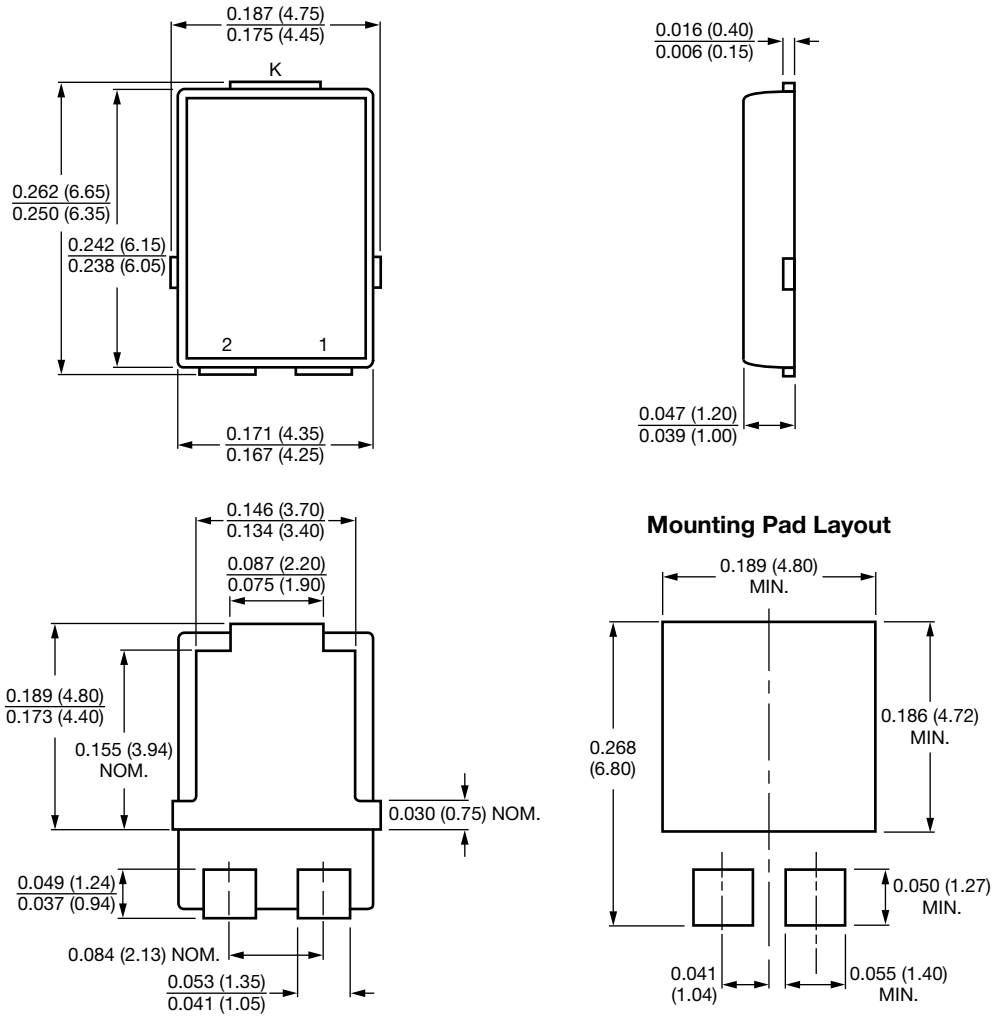


Fig. 6 - Typical Transient Thermal Impedance

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PACKAGE OUTLINE DIMENSIONS in inches (millimeters)

SMPC (TO-277A)



Mounting Pad Layout

Conform to JEDEC® TO-277A

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